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Hybrid QLSTM Framework for Time Series Forecasting in Dynamic Financial Markets

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***Abstract*—Prediction of stock price remains a challenging task because financial markets are dynamic and often non-linear, unpredictable patterns. To deal with such unpredictable data, the architecture that's created must identify both time based trends and hidden patterns in data. In this study, we explore a hybrid modelling approach by combining both the classical deep learning techniques with quantum computing principles. Specifically, a Long Short-Term Memory (LSTM) is combined with a Quantum Neural Network (QNN) to understand the benefits of using quantum in time series forecasting. Rather than depending on a single quantum configuration, multiple feature maps and ansatz circuit designs were evaluated to identify the most suitable arrangement for the QLSTM model. Resultant framework blends ideas from both classical computing and quantum approaches to improve how we predict patterns over time in real-life scenarios.**

***Index Terms*—QLSTM Framework, Quantum Computing, Time Series Forecasting, Stock Market Prediction, QML.**

I. Introduction

Predicting time series data is hard due to markets uncertain, non-linear and unstable nature. In sectors like trading, energy, and finance, where small improvements can have major impact, accurate forecasting is necessary. LSTM models are used often to deal with these problems,However, Long Short-Term Memory (LSTM) is well suited for sequential data [3], but they require lot of data, may miss hidden patterns and struggle when the data is limited or noisy and often fail to capture the hidden correlation in complex patterns [4].

To overcome the Challenges in LSTM, Quantum Computing offers new approaches, enabling quantum computers perform calculations in parallel within complex quantum states and richer data representations, potentially improving prediction accuracy with fewer parameters [1], [9]. Quantum Neural Network (QNN) are a key technology in this area which uses special quantum circuits to encode classical input data to process data in a interesting new ways [14]. To utilize both quantum and classical model strengths Hybrid QLSTM is used, The Hybrid Quantum Long Short-Term Memory (QLSTM) frameworks combine classical LSTM memory with quantum systems, shares promise for handling scarce or inconsistent data using variational quantum circuits and optimization technique. These frameworks are well suited for NISQ machines because they use circuits with components such as RealAmplitudes and ZZFeatureMap [13], [12], [11].

Some real world applications where QLSTM is used are stock market trends [6] energy forecasting for solar power systems [12] and finance for pattern recognition and analytics [5]. It leverages quantum properties to enhance learning, reduce overfitting and improved generalization [7], [8].In this work, The proposed hybrid framework architecture uses LSTM followed by Quantum Neural Network and classical output layer aiming to leverage both temporal trading and quantum computing power for improved financial stock price prediction [14], [10].

II. Related Studies

Over the years in the sectors such as finance, energy, and risk management the Long Short Term Memory (LSTM) networks plays a crucial role for High accuracy predictions to support strategic choices. However, LSTM need large dataset and often face issues with vanishing gradients in long sequences and they generalize poorly in high dimensional or noisy conditions.

To overcome these challenges, researchers actively examined on hybrid quantum-classical architectures, Utilizing Quantum computing ability of efficient information processing across high dimensional Hilbert spaces. Among many different techniques, Quantum Long Short Term Memory networks offer enhanced training efficiency while maintaining lowers architectural complexity.

A notable example is, Gong *et al.* [4], integrated Quantum Particle Swarm Optimization (QPSO) an optimization strategy with CNN-LSTM architecture to improve Optimization speed

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and to avoid local minima in electricity price forecasting.The results indicates improved optimization and higher prediction accuracy in unstable market environments highlighting optimization stratergies for real time financial applications.The comparative study by Saha *et al.* [3] showed that the QLSTM model outperformed classical LSTM model on the time series task, where the QLSTM model only required a fewer trainable parameters and attaining better performance with less training data.

Several studies applied quantum computing principle to attention mechanisms, aiming to utilize quantum parallelism and entanglement to enhance their ability to capture long time correlations to increase the learning capacity in deep learning frameworks. In their study, Lokes *et al.* [5] introduced a quantum self attention mechanism that used hybrid encoding strategies for financial time series forecasting enabling the model precisely capture temporal features in unpredictable market conditions. Meanwhile, Hsu *et al.* [7] implemented a quantum kernel based LSTM network for climate time series data which demonstrated the quantum kernels are capable of identifying complex non-linear relationships that traditional approached fail to represent.Early investigations by Mahajan [8] into Quantum Neural Networks (QNNs) which is used for predicting stock prices by offering the theoretical foundation for more complex hybrid models. In the comprehensive analysis Srivastava *et al.* [9] evaluated several quantum machine learning models for stock prediction that revealed the crucial role of quantum encoding strategies and entanglement mechanism which greatly enhanced model performance.While, Choudhary *et al.* [11] introduced HQNN-FSP a hybrid quantum neural network which is designed for regression tasks in financial markets achieving best performance.

Beyond finance, researchers have made notable strides in applying quantum computing to various domains. For example, Kea *et al.* [13] investigated on quantum-enhanced Long Short-Term Memory (LSTM) networks for predicting stock prices. Meanwhile, Tsurkan *et al.* [16] developed the Hybrid Quantum Recurrent Neural Networks to estimate the Remaining Useful Life (RUL) in industrial systems. Some other notable contributions which includes Moon *et al.* [17] QSegRNN for time series segmentation and *et al.* [18] Multi-Parallel QLSTM for modeling complex spatial-temporal relationships.

Researchers have made significant progress on applying quantum computing to the real-world problems. For example,Ubale *et al.* [19] successfully used hybrid Quantum Long Short-Term Memory (QLSTM) networks for fraud detection and achieved early detection accuracy and model responsiveness. Similarly, de Jesus *et al.* [20] explored that the Quantum Neural Networks (QNN) for demand forecasting concluded that the hybrid quantum-classical models performed well even with less data. Hsu *et al.* [21] supports that these findings with their study on quantum kernel-based LSTM by highlighting the potential of the quantum computing in sequential learning.

Despite progress, challenges persist due to Limited versatility, Practical hurdles (circuit depth, stability, hardware), Need for comparisons with classical models, Impact of quantum layers needs further study, These challenges hinder widespread adoption and effectiveness.

This study proposes a hybrid QLSTM framework for financial time series forecasting which aims to Minimize parameters, Maximize learning efficiency, Benchmark against classical and quantum models. The goal is to create a robust and practical model for real world forecasting.

III. Methodology

A. Raw Data

The Yahoo Finance via yfinance Python library, which provides historical stock data for the AAPL ticker, was used to access the Apple Inc. data. As shown in Fig.[I], with weekends and market holidays omitted, the data covers approximately six years from June 3, 2019, to May 30, 2025 and includes 1,508 daily observations. One record is sufficient for stock price forecasting because it contains essential stock market information such as the opening price, day's high and low prices, closing price, adjusted closing price, and trading volume.

Start Date	2019-06-03
End Date	2025-05-30
Total Records	1508
Training Samples	1206
Testing Samples	302

TABLE I: Raw Data

B. Exploratory Data Analysis (EDA)

The depth analysis of the stock data of Apple Inc. is done to recognize patterns, linkages, and statistical correlations between the prediction features

1) Closing Price Chart Over Time: With little change in prices brought about by market events, a graph of Apple's daily closing price indicated that the stock would likely continue to rise between 2019 and 2025. As shown in Fig. [1], this trend made it logical to examine price changes using time-based models such as LSTM.

Fig. 1: losing Price Over Time

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2) Moving Averages (MA10, MA50): The 10 and 50 day moving averages show's how the stock price moves over time. The Fig [2] shows that these averages reduces the impact of daily price changes and helps us to see clearer patterns and trends in the closing price.

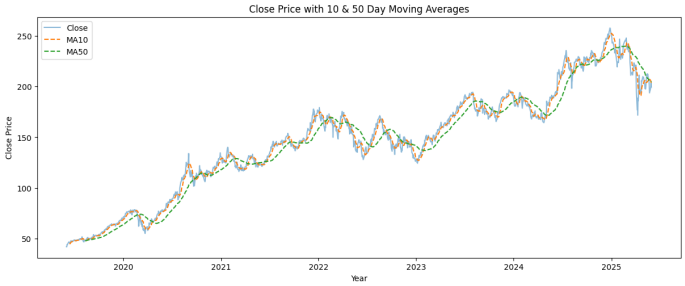

Fig. 2: Moving averages (MA10,MA50)

3) Yearly Trading Volume Comparison: The Fig. [3] shows that the volume peaked in 2020 during the Covid-19 crisis and it slowly dropped in later years. As the data is available till May so the volume is low in 2025.

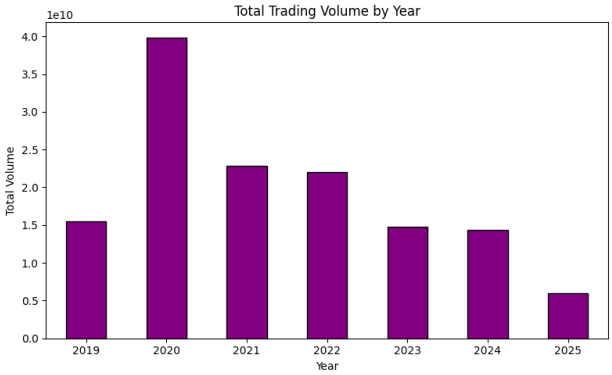

Fig. 3: Total Trading Volume by Year

4) Quarterly OHLC Price Trends: The average of Open, High, Low and Close prices for each quarter is calculated to spot longer term trends more clearly. The Fig. [4], shows that the Apple's stock generally moves upward from 2019 to early 2025, and it faces a small drops in 2022 and 2024. The close space between the price lines shows the steady movement within each quarter makes the view easier to understand for spotting trends over time.

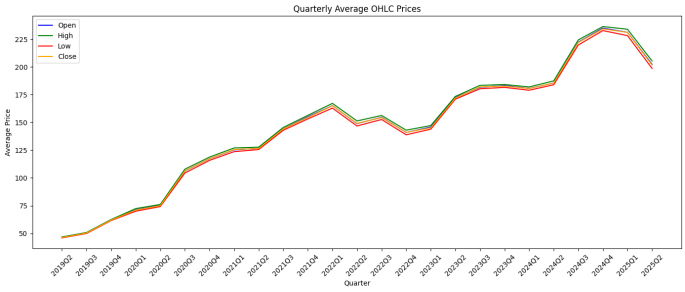

Fig. 4: Quarterly OHLC Price Trends

5) Days with Gains vs Losses: From the Fig [5] the chart explains the number of days with stock gain and losses over the period,it shows us that the gaining days are over 800 comparing to the losing days of 680. This suggest that it is a positive market trend where the stock moves upwards more frequently than declines.

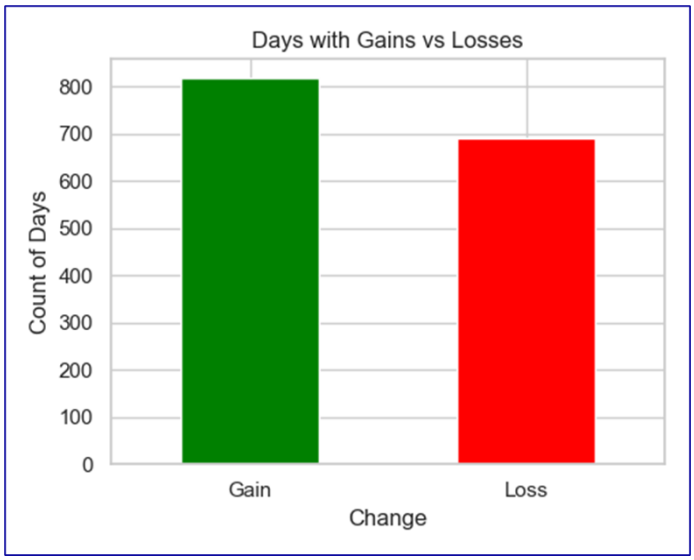

Fig. 5: Days with Gains vs Losses

C. Data Preprocessing

Selecting the essential columns from the dataset: Open, High, Low, Close, and Volume. Any rows with missing or non-numeric values were removed to ensure no missing or broken parts that could affect model training.

The values then normalized between 0 and 1 to make the process easy for the model. The Close price served as the target so it was excluded from the input features.

A sliding window approach is handled the model was given 20 days input and asked to predict price for the next day. It helped the model to capture short-term temporal patterns effectively. To prevent data leakage the dataset was split chronologically.

Data were divided into 48% of the data for training, 32% for validation, and the remaining 20% for testing as illustrated in Fig. [6].

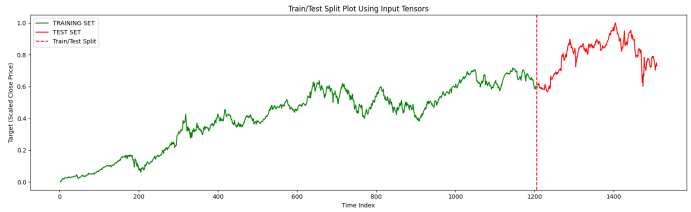

Fig. 6: Train Validation Test Split Of Target

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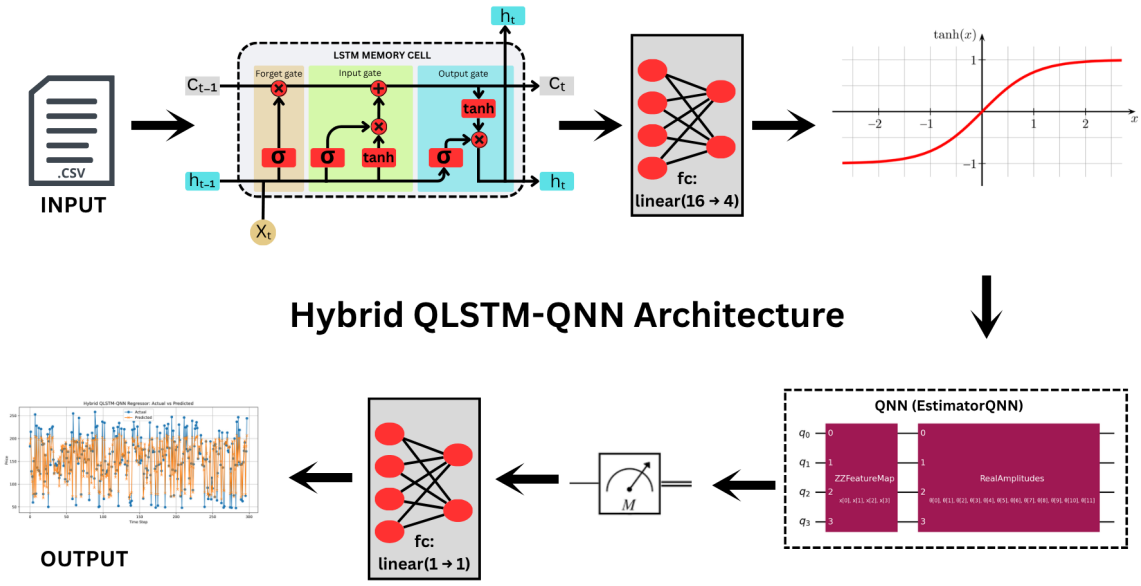

Fig. 7: Hybrid QLSTM-QNN Architecture used for stock price prediction.

D. Quantum-Classical Hybrid Model Architecture

As shown in Fig.[7], The model combines regular deep learning with quantum computing. It learns patterns over time in stock prices that which is used by the quantum circuits to better understand the data which helps improve prediction so that the model generalizes instead of memorizing

1) Classical LSTM: The 20 days of past stock data which includes Open, High, Low and Volume prices are used by the LSTM layer, shown in Fig.[8]. This layer is good at finding patterns and relationship over time. The hidden output summarizes the behavior of the stock during the period , it helps the model to make better predictions.

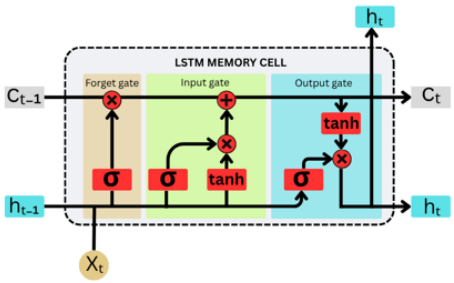

Fig. 8: Classical LSTM architecture inspired by [15].

2) LSTM to Quantum Layer: The dense layer is used to reduce the output it receives form LSTM. It projects the LSTM hidden state of 16 to a four dimensional vector matching the number of qubits in the quantum circuit. To make a meaningful quantum encoding (`tanh`) activation is applied to the input values to the range [-1,1].

3) Quantum Neural Network (QNN) Layer: The four values that the quantum layer received from the dense layer is encoded by the quantum circuit using a ZZFeatureMap with two repetitions as shown in Fig.[9]. It converts the input in the format that quantum circuit can utilize. The RealAmplitude which is of two layer is then employed with a full entanglement. The result is a flexible quantum circuit that identifies the complex patterns form the data that classical models struggles to do.

4) Forward and Backward Pass in QLSTM-QNN: The QLSTM architecture process data in two main stages they are forward pass and backpropagation. In the forward process, a classical LSTM first learns patterns from the input. The result is then reduced using a dense layer, and a tanh function keeps the values between -1 and 1. This is then given to the Quantum Neural Network (QNN).As shown in Fig. [10], The QNN with the help of featuremap and ansatz circuits encodes input into a quantum state and applies entanglement and quantum rotations. This allows the model to explore Hilbert space, that the classical model can't capture. After the the output is received form the quantum layer, The classical layer process the final prediction.

In the backpropagation, Usage of hybrid optimization strategy helps the classical and quantum parameters to compute the gradients, the trainable quantum gate angles and the classical weights are updated with the help of this strategy. This architecture of quantum computing helps enhancing the model's expressiveness and learning capacity, especially for capturing nonlinear dependencies in time series data

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E. Quantum Featuremap and Ansatz

1) Feature Map: It's is a parameterized quantum circuit that encodes data into quantum state, this maps the data into the Hilbert space using quantum gates, including rotations (e.g., `RX`, `RZ`) and entangling operations such as `CNOT` gates. The transformation helps make complex data easier to separate by mapping it into a quantum space.

Mathematically, for a given classical input vector $x \in \mathbb{R}^n$, a quantum feature map $\Phi(x)$ embeds it into a quantum state $|\Phi(x)\rangle$ such that:

$$\Phi : \mathbb{R}^n \to \mathcal{H}, \quad x \mapsto |\Phi(x)\rangle$$

where $\mathcal{H}$ is a complex Hilbert space.

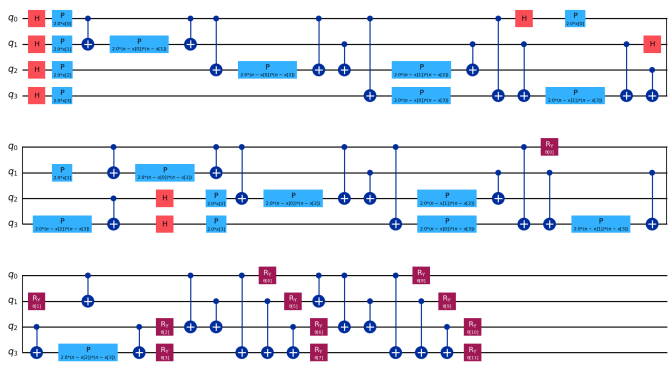

Fig. 9: Featuemap and Ansatz circuit

2) Ansatz: It's a part of quantum circuit that learns patterns form the data it's achieved with the help of rotations and entanglements. The ansatz explores the Hilbert space and provides expressivity which is required to learn the complex patterns . It's similar to how weights are learned in classical neural network. Without ansatz only the encoding operation is performed no learning happens. With an ansatz, the model becomes a trainable function $f(x;\theta)$ that can perform tasks like classification or regression. Mathematically, it is expressed as:

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

where:

$U(\theta)$ is a parametrized quantum circuit (sequence of gates) with trainable parameters θ. $|\psi\rangle$ is the resulting quantum state. $|0\rangle$ is the initial state (usually all qubits in the $|0\rangle$ state).

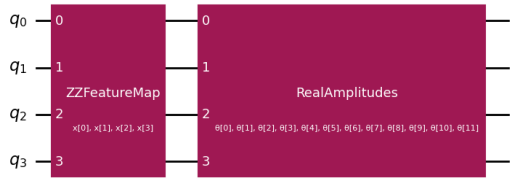

Fig. 10: Featuremap Ansatz Architecture

IV. Results and Discussion

The hybrid QLSTM architecture with combined classical LSTM network with a quantum neural network takes in input sequence of financial features like Open, High, Low, Volume over a fixed window (sequence length =20) . The first layer, LSTM captures temporal dependencies and outputs a hidden representation at a final timestep. This output is sent through a fully connected layer followed by a tanh activation to make it in a format that is suitable for the quantum layer to process. The resultant vector is fed to a Quantum Neural Network, which uses a quantum circuit made by Featuremap and Ansatz and 4 qubits, because of the usage of 4 features to predict Close price. The QNN transforms the data to find the non linearity , with the help of Hilbert space and the output is passed to final dense layer to predict final close price. The model is trained with the help of RMSprop optimizer and MSE loss, with enabling early stopping shown in Fig. [11]

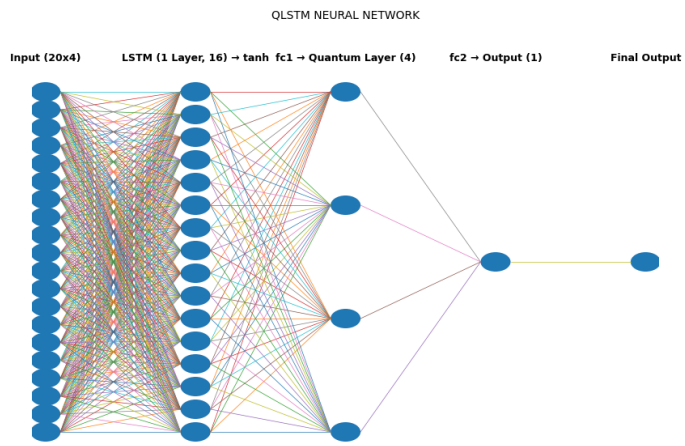

Fig. 11: QLSTM Hybrid Neural Network Architecture

A. Finalizing Quantum Circuit Design

For selecting the best Quantum circuit design different combination of Featuremaps and Ansatz circuit were tested in the QLSTM's QNN layer. The evaluation was based on the metrices like MSE, MAE, RMSE, R² and Explained Variance. On the evaluation process the Quantum circuit with combination of ZZFeaturemap and Realamplitude , PauliFeatureMap and RealAmplitude performed well compared to other quantum circuits. Both achieved very low errors (MSE 0.0018) and high R² values (0.83), where other Quantum circuit showed higher errors and even negative R2 , indicating unstable predictions. This experiment helped determine the optimal quantum circuit for the QLSTM framework.

Type	MSE	MAE	RMSE	R^2	Explained Variance
ZZFeatureMap & RealAmplitudes	0.0018	0.0336	0.0424	0.8372	0.897
PauliFeatureMap & RealAmplitudes	0.0018	0.0338	0.0424	0.8347	0.8932
PauliFeatureMap & EfficientSU2	0.1186	0.3406	0.3444	-9.9449	0.7589
ZZFeatureMap & EfficientSU2	0.2498	0.4921	0.4998	-22.0447	0.2913
ZFeatureMap & RealAmplitudes	0.3577	0.5889	0.5981	-31.9975	-0.0019
ZFeatureMap & EfficientSU2	0.5023	0.701	0.7087	-45.3378	-0.005

TABLE II: Performance comparison of different quantum feature map and ansatz combinations

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B. Comparison of LSTM and QLSTM Models

In the Table. [III], The evaluation of the LSTM and Quantum enhanced QLSM model is based on the metrices like MSE, MAE, RMSE, R2 and Explained Variance. The QLSTM model with 4 qubits , 100 training epochs, a hidden size of 16, a learning rate of 0.01, and the RMSprop optimizer achieved the best overall performance, this model attained a R^2 score of 0.83 and the lowest MSE of 85.16, along with the lowest MAE (7.49) and RMSE (9.23), indicating that the model is superior in accuracy and generalization capability. But the classical LSTM yielded R^2 of 0.54 and higher error values. Finally the results demonstrated that the Quantum models which is appropriately tuned can outperform the classical counterparts in time series forecasting tasks.

Model name	Qubit No.	Epochs	Hidden Size	Learnig Rate	Optimizer	R2	MSE	MAE	RMSE	Explained Variance
LSTM	-nil-	100	16	0.01	RMSprop	0.54	235.74	13.53	15.35	0.87
QLSTM	5	100	32	0.01	Adam	-0.08	552.52	20.68	23.51	0.70
QLSTM	4	100	16	0.1	RMSprop	0.50	252.68	13.32	15.90	0.82
QLSTM	4	100	16	0.01	RMSprop	0.83	85.16	7.49	9.23	0.89
QLSTM	4	100	16	0.001	RMSprop	0.76	120.85	9.13	10.99	0.88

TABLE III: QLSTM and LSTM comparison table

C. Loss Convergence

As shown in the Fig [12], The QLSTM's smooth loss curve indicate that quantum layer contributes a lot to capture complex pattern efficiently. The model after a few epoch shows a fast and smooth drop in both training and validation losses and the losses become almost zero. This shows that the QLSTM model learns faster and generalizes better to new data.

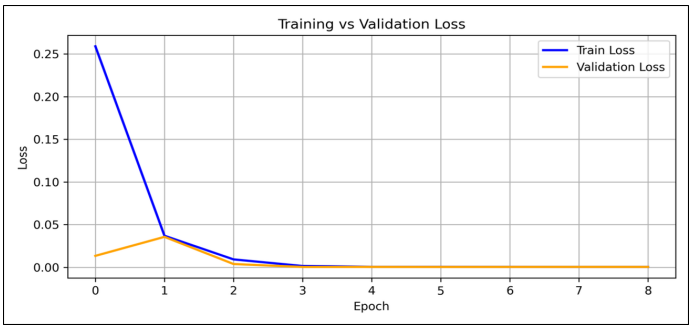

Fig. 12: QLSTM Training vs Validation Loss

The fig [13] shows the classical LSTM model, struggles to generalize and overfits or not stable during training. The training loss gradually decreases but validation loss shows large ups and downs. This shows that the model is memorizing instead of generalizing and the model is not performing well on unseen but learns on training data

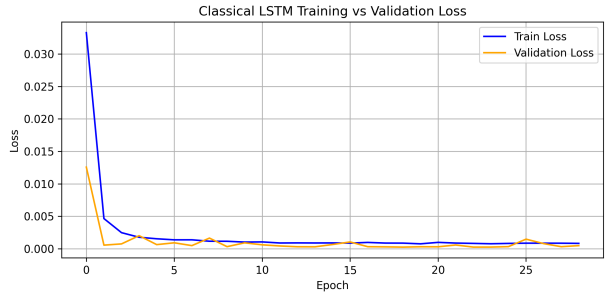

Fig. 13: Classical LSTM Training vs Validation Loss

D. Model Comparison: QLSTM vs LSTM

The Fig [14], the actual vs predicted plot show's that the QLSTM model with 4 qubits , a hidden size of 16, learning rate of 0.01 and RMSprop optimizer tracks the price movements closely, including rapid fluctuations, suggesting better generalization. The evaluation of QLSTM shows us that it achieves highest R^2 of 0.83, MSE of 85.16, MAE of 7.49, RMSE of 9.23 and explained variance of 0.89. It shows that the hybrid model's capability to capture complex non linear trends providing better generalization.

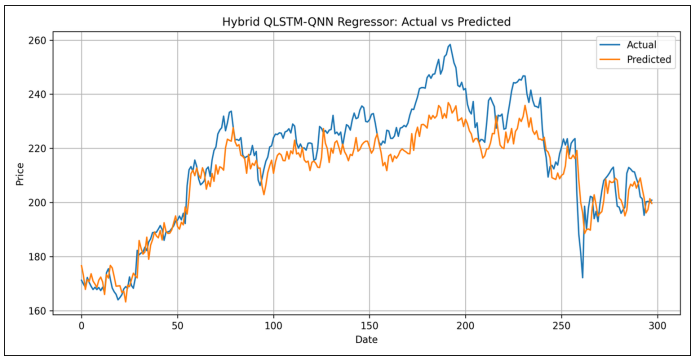

Fig. 14: QLSTM Actual vs Predicted

The Fig [15] shows that the classical LSTM model memorizes the data instead of generalizing and fails to capture spikes or drops in the actual data. But the LSTM model follow the general trend but miss to follow up in the regions with sharp price movements. In the evaluation of LSTM shows the moderate prediction accuracy, R^2 of 0.54, MSE of 235.74, MAE of 13.53 and RMSE of 15.35 with higher error margins.

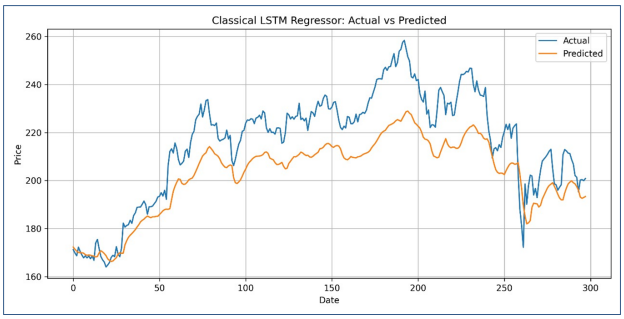

Fig. 15: Classical LSTM Actual vs Predicted

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V. Conclusion

This framework shows that Stock Market Data which is a complex time series data, can be handled by combining a classical LSTM with an efficient use quantum neural network to accurately predict the required field. It demonstrates, by integrating a classical LSTM with Quantum layer comprising ZZFeatureMap and RealAmplitude layers enable accurate prediction of complex time series data like stock prices. The configuration of quantum circuit was chosen after testing several quantum circuit combinations. It consistently provided the lowest error values and more reliable predictions showing its effectiveness in capturing nonlinear patterns in financial data. The integration of quantum layer to a classical model improves the performance and it provides new approaches to handle complex data using quantum machine learning techniques.

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